MATH 4511 – Assignment 2

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The solutions are handwritten and included as scans. The numerical answers are computed in Python (notebook `MATH4511_assignment2.ipynb`); each code block below is copied from that notebook and is followed by its output.

Problem 1 – Pricing an on-going interest rate swap

The swap is valued from the position of the floating rate receiver (fixed rate payer), $V = V_{\text{floating}} - V_{\text{fixed}}$. The remaining payment dates are $t_1 = 3/12$, $t_2 = 9/12$ and $t_3 = 15/12$, and the spot rates (nominal, compounded semi-annually) give the discount factors $B(0;t) = (1 + r_t/2)^{-2t}$.

Assignment 2 Math 4511

Exercise 1

a) $V_{swap} = V_{floating} - V_{fixed}$

$V_{fixed} = \sum_{t=1}^{k=6} N \cdot \frac{R_{fixed}}{2} B(0, \frac{6t-3}{12})$

$V_{floating} = N R_{last} B(0, \frac{3}{12}) + \sum_{t=1}^{2} N\left(B(0, \frac{6(t-1)}{12} + \frac{3}{12}) - B(0, \frac{6t}{12} + \frac{3}{12})\right)$

$R_{floating} = R[\frac{1}{2}, \frac{3}{2}]$

$= N R_{last} B(0, 3/12) + N\left(B(0, 3/12) - B(0, \frac{15}{12})\right)$

$V_{swap} = N\left(\frac{R_{last}}{2} B(0, 3/12) + B(0, 3/12) - B(0, 15/12) + \left(\sum_{t=1}^{3} \frac{R_{fixed}}{2} B(0, \frac{6t-3}{12})\right)\right)$

$V_{swap} = 0.0034\, N$

b) We can price the swap by considering the difference in value between a floating rate note and fixed rate bond

$V P_{Bond} = \sum_{t=1}^{3} \frac{C}{m} F \cdot B(0, \frac{6t-3}{12}) + F B(0, 15/12)$

where $F = N$ is the face value, m is the annual payment frequency which is two, and $C = r_{fixed}$ is the coupon rate

$P_{frn} = \left(F \frac{r_{last}}{2} + F\right) B(0, 3/12)$

$\Rightarrow V_{swap} = F\left(\left(\frac{r_{last}}{2} + 1\right) B(0, 3/12) - \left(\sum_{t=1}^{3} \frac{r_{fixed}}{2} B(0, \frac{6t-3}{12}) + B(0, 15/12)\right)\right)$

Assignment 2 MATH 4511
Exercise 1

c) We can also use forward rates the price the floating cash flows.

We use the following relationship

$$(1+r_{t_1})^{t_1}(1+f_{t_1,t_2}) = (1+r_{t_2})^{t_2}$$

$$\Rightarrow f_{t_1,t_2} = \frac{(1+r_{t_2})^{t_2}}{(1+r_{t_1})^{t_1}} - 1 = \frac{B(0,t_1)}{B(0,t_2)} - 1$$

where f_{t_1,t_2} is the effective rate between the payment dates

If we let $t_1 = 3/12$, $t_2 = 9/12$ and $t_3 = 15/12$, then

$$V_{floating} = \frac{r_{last}}{2} B(0,t_1) + \left(\frac{B(0,t_1)}{B(0,t_2)} - 1\right) B(0,t_2) + \left(\frac{B(0,t_2)}{B(0,t_3)} - 1\right) B(0,t_3)$$

floating payment 1 — floating payment 2

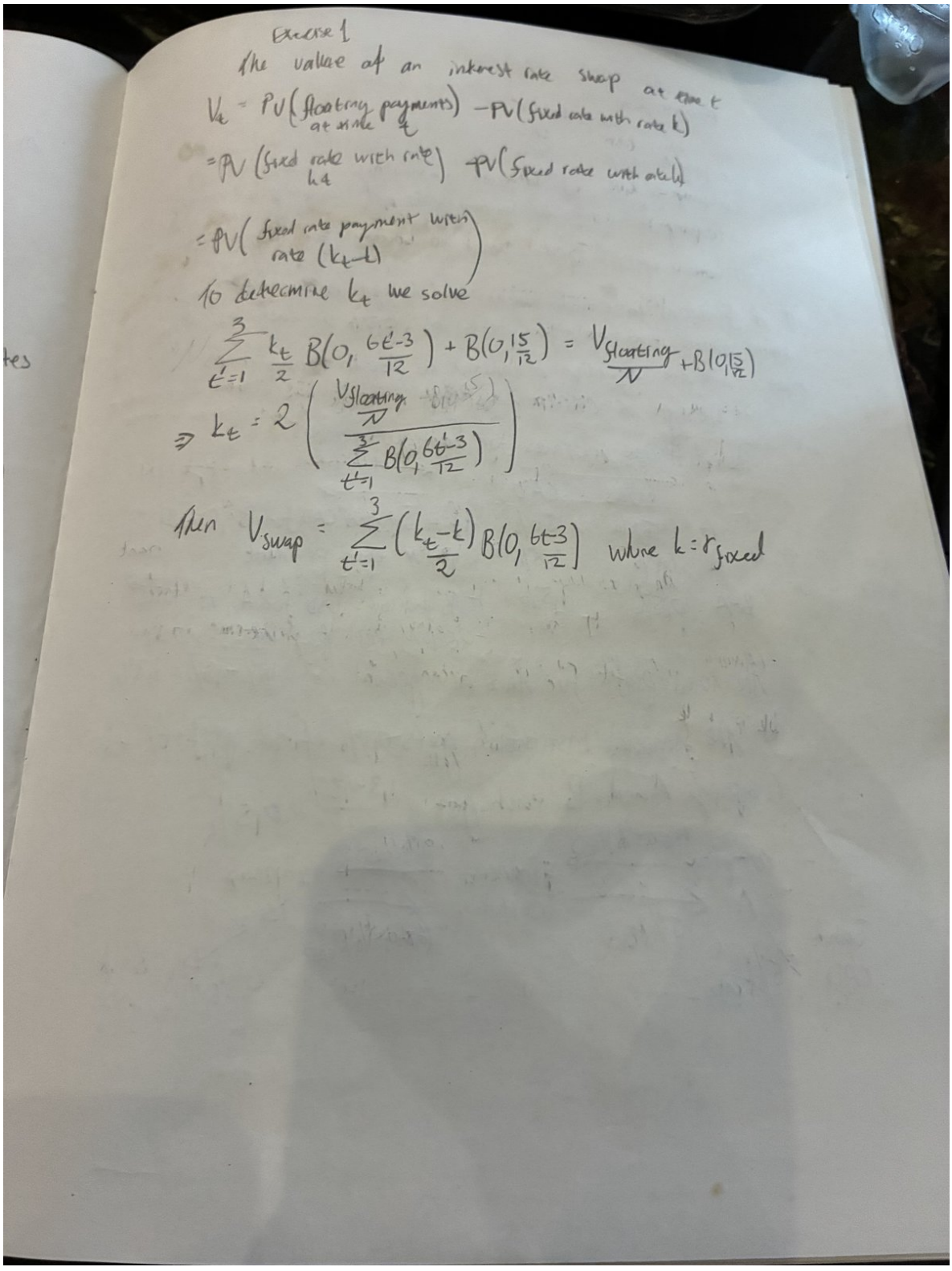
For the fixed leg we can use the fixed rate bond method.

The value of the swap is equal to the difference in values between the floating and fixed leg.

d) Swap rate of the same swap quoted at current time t

When the swap is created the value of the fixed and floatings legs are the same. This is achieved by setting the fixed rate k_t to the appropriate amount that the payments match the discounted cash flows of multiple floating rate agreements (or long-short bonds with different maturities)

then PV(floating leg at time t) = PV(fixed leg payment with coupon rate k_t)




```
spot_rates = { # spot rates quoted as annual nominal rates
    3/12: 0.034,
    6/12: 0.035,
    9/12: 0.036,
    12/12: 0.037,
    15/12: 0.038
}
# the exchanges are made semiannually in the swap and the next exchange will be made 3 months from
    today
m = 2
```

```
r_fixed = 0.036
r_last = 0.037 # the 6 month annual nominal spot rate the period before
# We are going to calculate the cost to enter the fixed rate position in the swap, ie the party
    that pays the fixed rate and received the floating rate
payment_times = [3/12, 9/12, 15/12]
```

```
def r_eff(r_nominal, m = 2): # converts rates to annual effective rates
    return (1 + r_nominal / m) ** m - 1
def B(spot_rates, t): # converts the spot rate to a discount factor
    spot_rate = spot_rates[t]
    effective_rate = r_eff(spot_rate)
    return (1 / (1 + effective_rate) ** t)
#a) Using the original formula
V_fixed = sum(
    r_fixed / 2 * B(spot_rates, t) for t in payment_times
)
V_floating = (r_last / 2)*B(spot_rates, 3/12) + B(spot_rates, 3/12) - B(spot_rates, 15/12)
V_swap = V_floating - V_fixed
print(f'The price of the swap agreement is ${V_swap:.4f}N, where N is the principal')

#b) Using the difference in value between a floating rate note and a fixed rate bond
P_bond = sum(
    r_fixed / m * B(spot_rates, t) for t in payment_times
) + B(spot_rates, payment_times[-1])
P_frn = (1 + r_last / m) * B(spot_rates, payment_times[0])
V_swap = P_frn - P_bond
print(f'The price of the swap agreement is ${V_swap:.4f}F, where F is the face value of the
    floating rate note and the fixed rate bond')

#c) Using forward rates

f_12 = B(spot_rates, payment_times[0]) / B(spot_rates, payment_times[1]) - 1
f_23 = B(spot_rates, payment_times[1]) / B(spot_rates, payment_times[2]) - 1
V_forward = (r_last / 2) * B(spot_rates, payment_times[0]) + f_12 * B(spot_rates, payment_times[1])
     + f_23 * B(spot_rates, payment_times[2])
V_swap = V_forward - V_fixed
print(f'The price of the swap agreement is ${V_swap:.4f}F, where F is the face value of the fixed
    rate bond and the principal of the forward rate agreement')

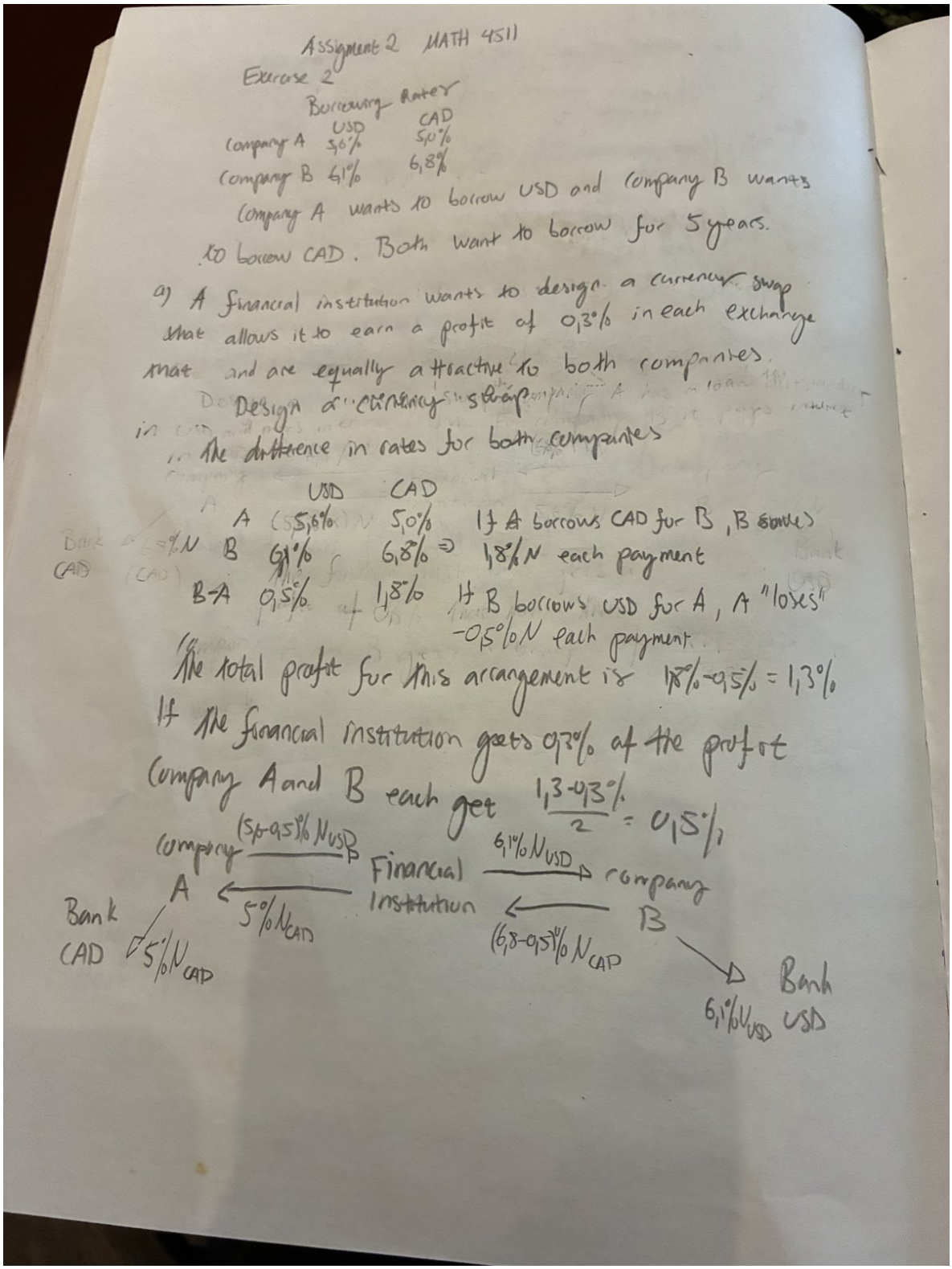
#d) Using swap rate quoted at current time
kt = 2 * (V_floating) / sum(
    B(spot_rates, t) for t in payment_times
)
V_swap = sum(
    (kt - r_fixed) / 2 * B(spot_rates, t) for t in payment_times
)
print(f'The price of the swap agreement is ${V_swap:.4f}N, where N is the principal of the swaps')
```

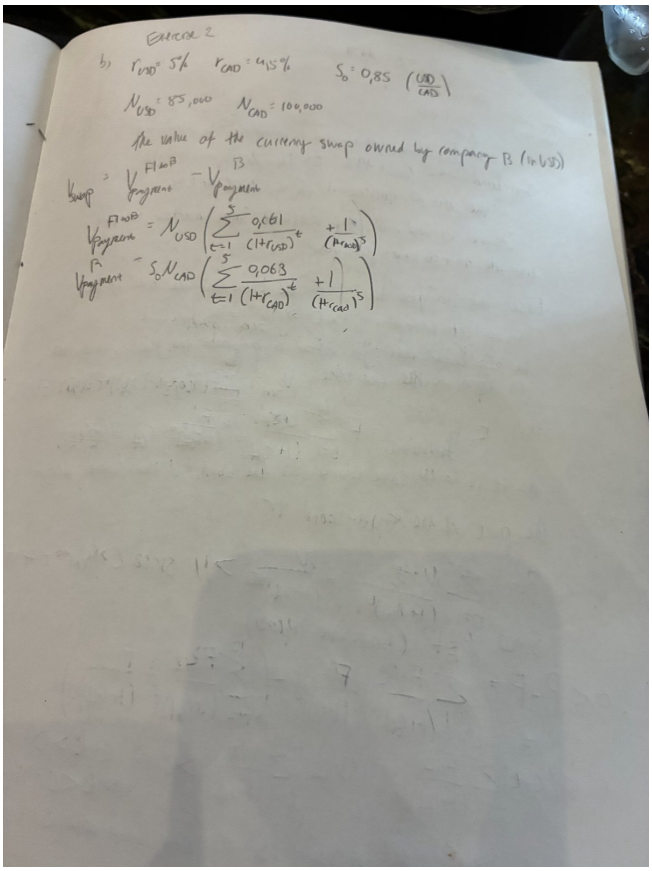
```
The price of the swap agreement is $0.0034N, where N is the principal
The price of the swap agreement is $0.0034F, where F is the face value of the floating rate note
    and the fixed rate bond
The price of the swap agreement is $0.0034F, where F is the face value of the fixed rate bond and
    the principal of the forward rate agreement
The price of the swap agreement is $0.0034N, where N is the principal of the swaps
```

Quantity	Value (per unit of N)
$B(0;3/12),\ B(0;9/12),\ B(0;15/12)$	0.991607, 0.973595, 0.954036
V_{fixed}	0.052546
V_{floating}	0.055916
Forward rates per half-year $f_{[t_1,t_2]},\ f_{[t_2,t_3]}$	0.018500, 0.020502
Swap rate today K_t (nominal)	3.831%

Answer: all four methods give $V \approx 0.003370\,N$. Since the swap rate today ($K_t = 3.831\%$) is above the fixed rate $K = 3.6\%$, the swap has positive value for the floating rate receiver.

Problem 2 – Currency swap





In the swap, A pays 5.1% on N_{USD} and receives 5.0% on N_{CAD}, and B receives 6.1% on N_{USD} and pays 6.3% on N_{CAD}. The principals are exchanged at time 0 and again at year 5. Both companies save 0.5% and the institution earns 1.3% in CAD minus 1.0% in USD, i.e. 0.3% at today's exchange rate; its profit is therefore exposed to exchange rate risk.
In (b) the rates are annual effective and the payments annual, so $B(0;t) = 1.05^{-t}$ (USD) and $B_F(0;t) = 1.045^{-t}$ (CAD). The CAD leg is converted to USD with $S_0 = 0.85$ USD per CAD.

```
#a)
r_FI_to_B = 0.061
r_B = 0.063

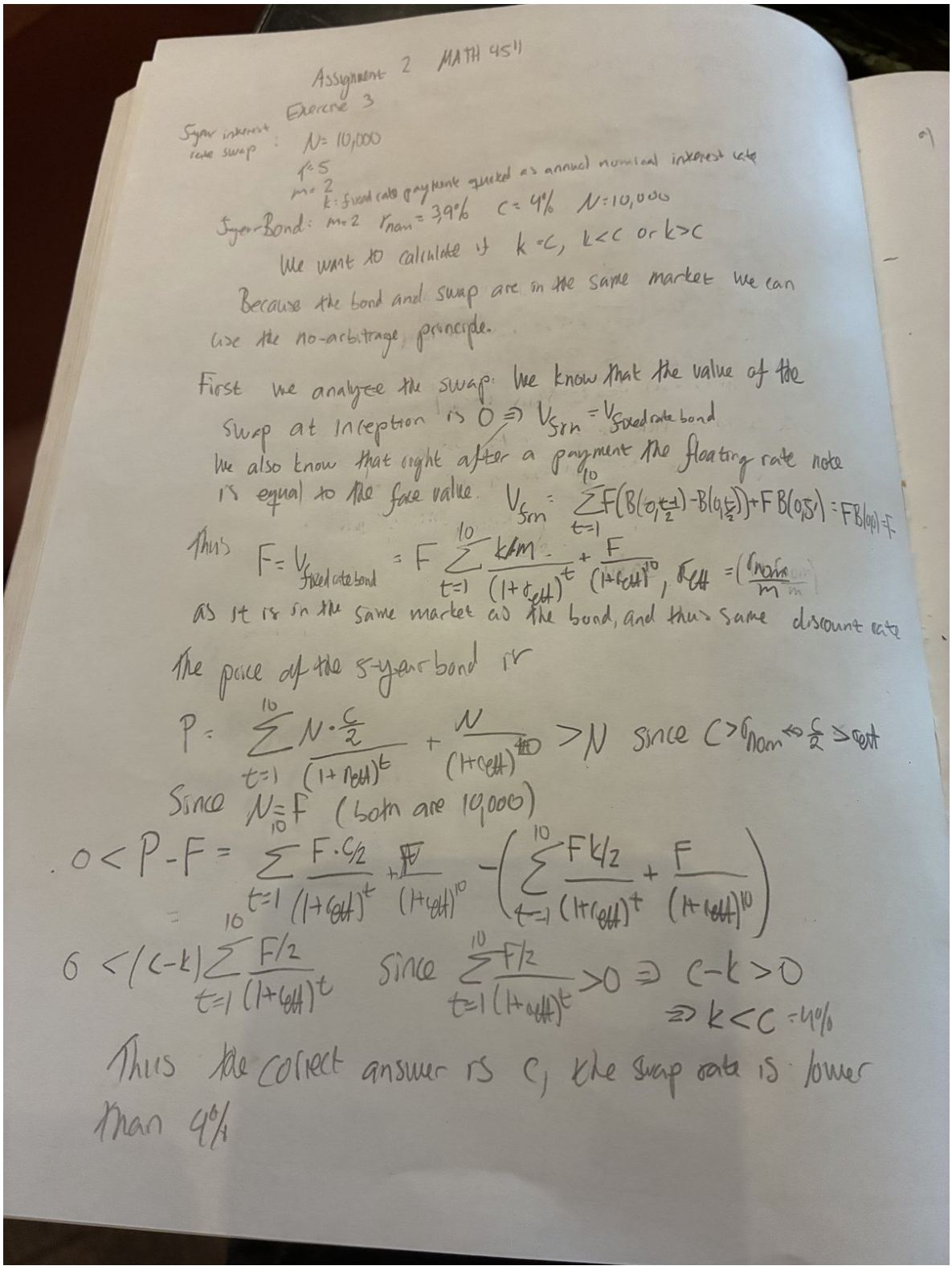
#b)
r_USD = 0.05
r_CAD = 0.045
N_USD = 85_000
N_CAD = 100_000
S = 0.85 # exchange rate

V_pay_in = N_USD*(sum(r_FI_to_B/(1 + r_USD) ** t for t in range(1, 5+1)) + 1/(1 + r_USD) ** 5)
V_pay_out = S * N_CAD*(sum(r_B/(1 + r_CAD) ** t for t in range(1, 5+1)) + 1/(1 + r_CAD) ** 5)
V_swap = V_pay_in - V_pay_out
print(f'The value of the currency swap for company B is ${V_swap:.4f}')
```

```
The value of the currency swap for company B is $-2668.6037
```

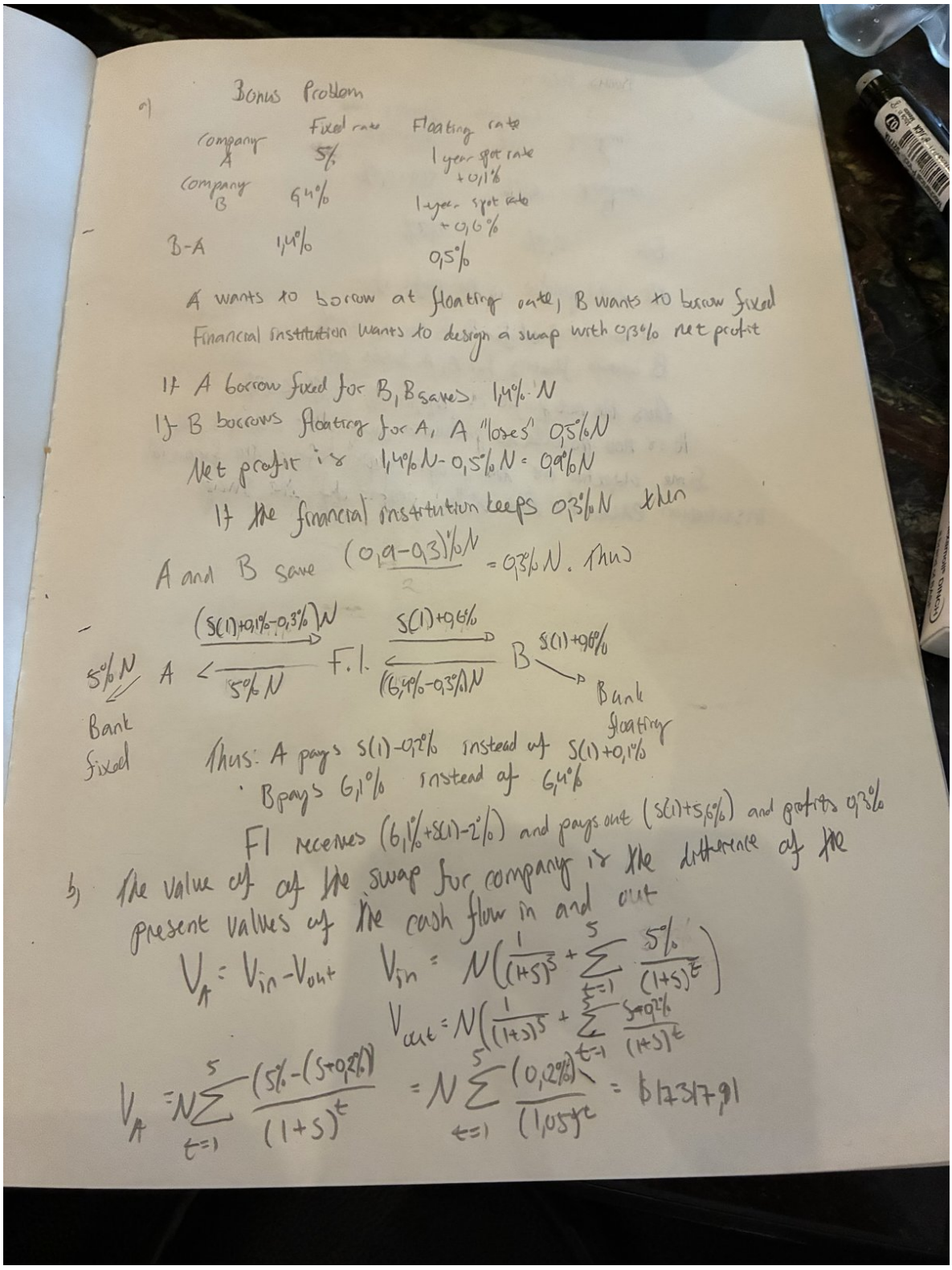
Answer: the swap owned by company B is worth about -2668.60 USD. B pays 6.3% in CAD while the CAD rate is only 4.5%, which outweighs the 6.1% it receives in USD against a USD rate of 5%.

Problem 3 – Swap rate versus coupon rate



With yield 3.9% the bond price is $P \approx 10\,045.03 > 10\,000$, so the bond trades above par. The swap rate is the coupon rate of a par bond, hence **(C)**: $K < 4\%$.

Bonus Problem – Interest rate swap



(b) Value of the swap owned by company A

In the swap of (a), A receives 5% fixed and pays $s(1) - 0.2\%$ to the institution. With a flat term structure at 5% all 1-year forward rates equal 5%, so the floating payments are valued with $s(1) = 5\%$ and discounted with $B(0;t) = 1.05^{-t}$. A notional principal is added to both legs; it cancels.

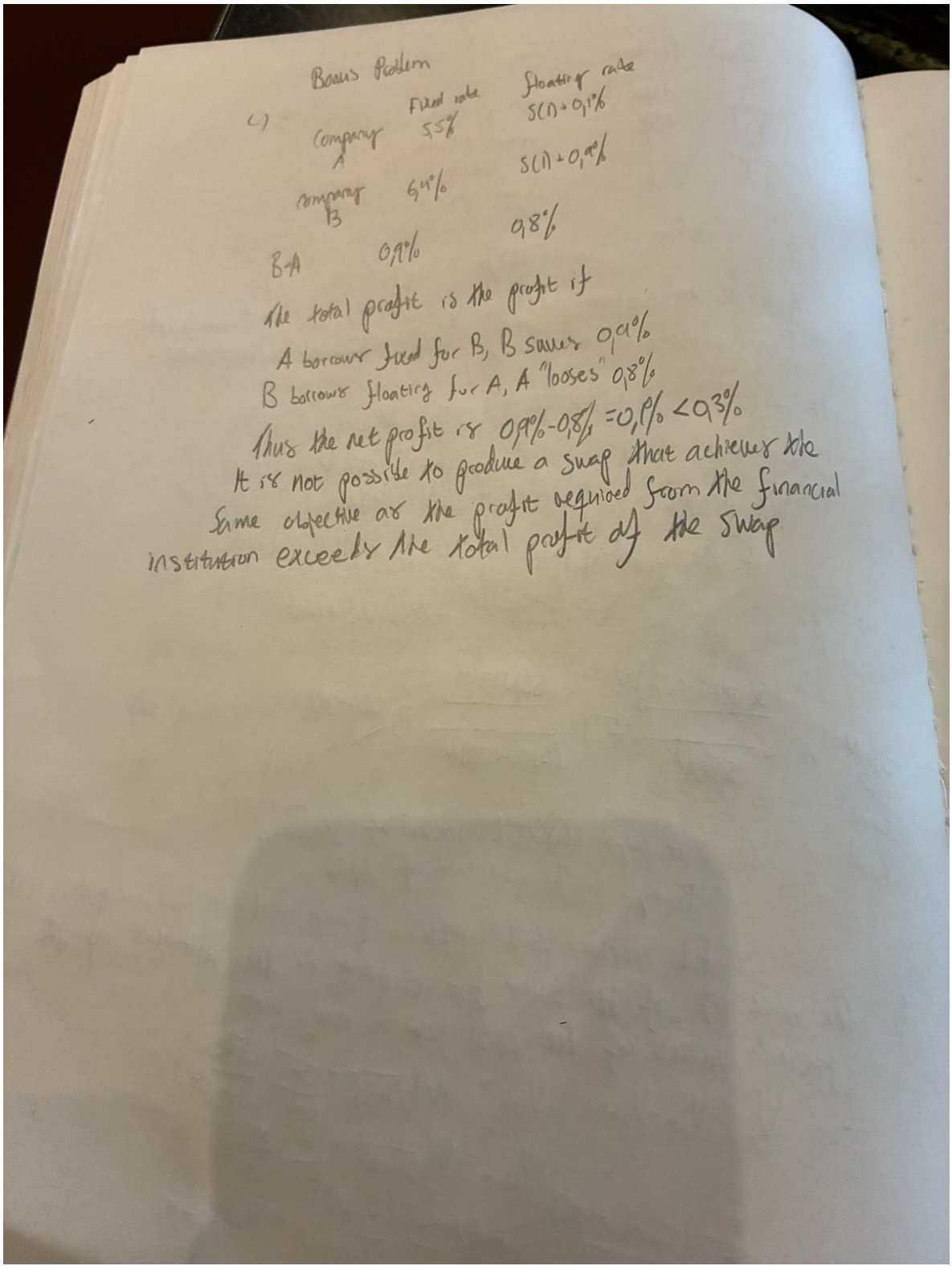
```python
#a) and b)
s1 = 0.05
N = 2_000_000
T = 5
A_rates = {
    "r_in": 0.05,
    "r_out": s1 - 0.002
}
B_rates = {
    "r_in": s1 + 0.006,
    "r_out": 0.061
}
FI_rates = {
    "r_in": s1 + 0.059,
    "r_out": s1 + 0.056
}

# The value for A is the difference between present value of the cash flows in vs the cash flows
    out
V_in = N * (1 / (1 + s1) ** T + sum(
    A_rates["r_in"] / (1 + s1) ** t for t in range(1, T + 1)
))
V_out = N * (1 / (1 + s1) ** T + sum(
    A_rates["r_out"] / (1 + s1) ** t for t in range(1, T + 1)
))
V_A = V_in - V_out
print(f'The value of the swap for company A is ${V_A:.4f}')
```

```
The value of the swap for company A is $17317.9067
```

Answer: $V_A \approx 17\,317.91$. The fixed leg is worth exactly N because its coupon equals the market rate, while A pays 0.2% less than the market rate on the floating leg.

(c)


Banks Problem
c)
Fixed rate
Floating rate
Company A
5,5%
S(1)+0,1%
company B
6,4%
S(1)+0,9%
B-A
0,9%
0,8%
the total profit is the profit if
A borrows fixed for B, B saves 0,9%
B borrows floating for A, A "looses" 0,8%
Thus the net profit is 0,9%-0,8% =0,1% <0,3%
It is not possible to produce a swap that achieves the
same objective as the profit required from the financial
institution exceeds the total profit of the swap